

# 2022 HKDSE Physics

## Paper 1B — Structured Questions

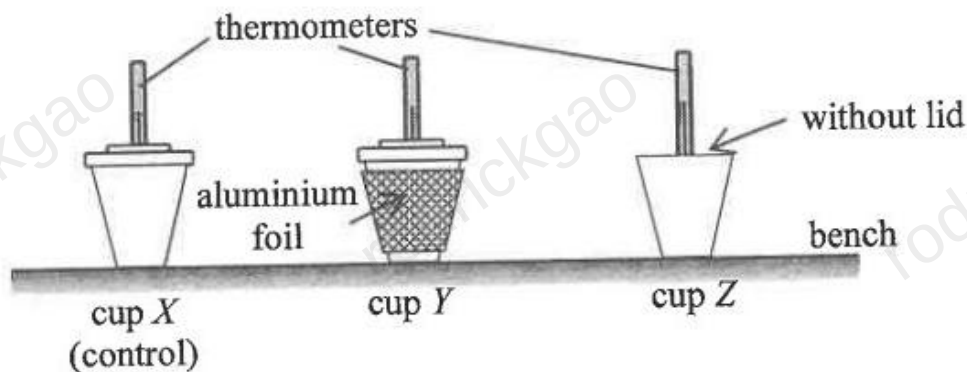
Original questions and official answers with detailed English reasoning. Calculations show the governing relationships, substitution and units; each structured subpart is presented separately.

**Editable questions with official answers and detailed English reasoning. Subscripts and superscripts remain editable.**

*Official answers are placed directly below each question.*

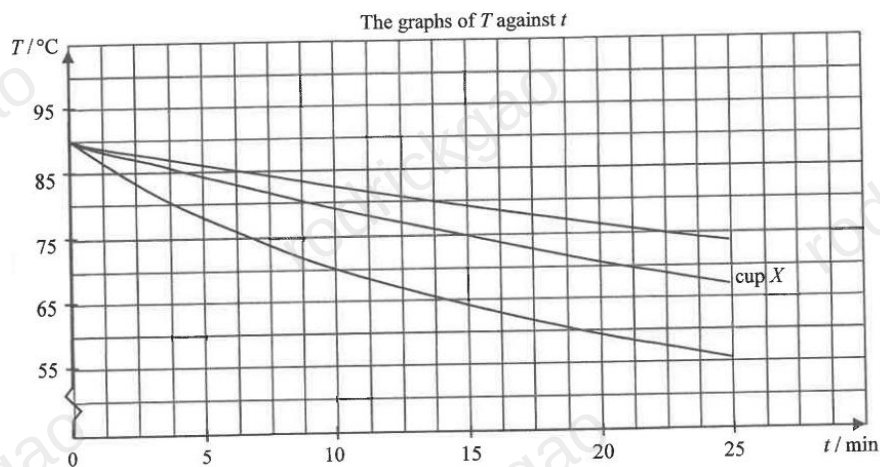
## Question 1 — Heat transfer

A student conducts an experiment to study how to best retain the warmth of water by using identical paper cups X, Y and Z shown in Figure 1.1. Each cup is filled with  $250 \text{ cm}^3$  of hot water and cup X serves as a control.



| Cup | Wrapping           | With a lid |
|-----|--------------------|------------|
| X   | no                 | yes        |
| Y   | aluminium wrapping | yes        |
| Z   | no                 | no         |

When the water temperature is  $90^\circ\text{C}$ , the student starts taking thermometer readings every minute. Figure 1.2 shows the variation of temperature  $T$  of water in the cups with time  $t$  elapsed.



- Suggest a reason why this experiment begins with the same initial water temperature ( $90^\circ\text{C}$ ). (1 mark)
- Explain why all the curves become less steep as time elapses. (2 marks)
- On Figure 1.2, indicate the respective curves that correspond to the results of cup Y and cup Z. (1 mark)
  - Explain your answer by referring respectively to the main heat transfer process. (3 marks)
- Suggest a material for making the cups so that heat loss by conduction through the cups can be reduced. (1 mark)

## Official marking scheme

## Paper 1 Section B

| Solution                                                                                                                                                                                                                                           | Marks | Remarks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------|
| 1. (a) Fair test (otherwise the initial temperature would affect the result).                                                                                                                                                                      | 1A    |         |
|                                                                                                                                                                                                                                                    | 1     |         |
| (b) Temperature difference between hot water and the surroundings (room temperature) decreases as time elapses. Rate of energy loss / heat loss becomes smaller, hence rate of temperature drop is smaller and the curve becomes less steep.       | 1A    |         |
|                                                                                                                                                                                                                                                    | 1A    |         |
|                                                                                                                                                                                                                                                    | 2     |         |
| (c) (i) The graphs of $T$ against $t$                                                                                                                                                                                                              |       |         |
|                                                                                                                                                                                                                                                    | 1A    |         |
|                                                                                                                                                                                                                                                    | 1     |         |
| (ii) Comparing to cup X as a control,<br>Cup Y:<br>The shiny surface of aluminium foil is a poor emitter / good reflector of radiation (thus reduces heat lost to surroundings).<br>Cup Z:<br>No lid allows convection through air / surroundings. | 1A    |         |
| Or<br>No lid enhances heat lost by evaporation / convection / contacting with air / surroundings.                                                                                                                                                  | 1A    |         |
|                                                                                                                                                                                                                                                    | 3     |         |
| (d) Polystyrene / foam plastic or foam / wood                                                                                                                                                                                                      | 1A    |         |
|                                                                                                                                                                                                                                                    | 1     |         |

(a) Use the same starting temperature for a fair comparison, because starting temperature itself affects the cooling rate.

(b) As the water cools, its temperature difference from the surroundings decreases. The rate of heat loss therefore falls, so the rate of temperature decrease becomes smaller and each curve becomes less steep.

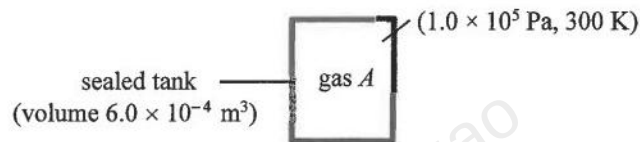
(c)(i) Label the upper curve Y and the lower curve Z; X is the middle control curve.

(c)(ii) Compare each cup with X. Y's shiny aluminium surface is a poor emitter of thermal radiation, reducing radiative heat loss. Z has no lid, allowing greater heat loss through evaporation and convection at the exposed water surface.

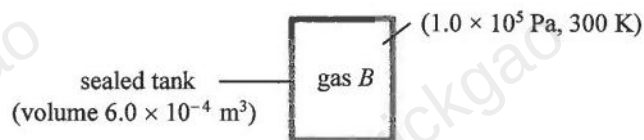
(d) Use polystyrene foam, which is a poor thermal conductor.

## Question 2 — Kinetic theory of gases

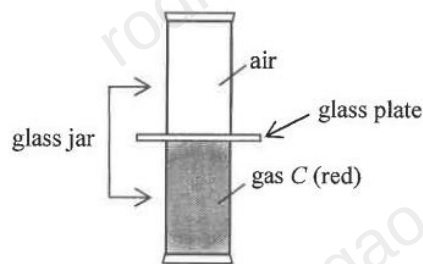
- \* (a) (i) Figure 2.1 shows a sealed tank of volume  $6.0 \times 10^{-4} \text{ m}^3$  containing a monatomic gas A at a pressure of  $1.0 \times 10^5 \text{ Pa}$  and at a temperature of  $300 \text{ K}$ .



- (I) Estimate the number of gas molecules in the tank,  $N$ . (2 marks)
- (II) Estimate the average kinetic energy of the gas molecules,  $E_k$ . (2 marks)
- (ii) Figure 2.2 shows another identical tank containing monatomic gas B under the same pressure and temperature. A molecule of gas B has one fifth the mass of a molecule of gas A.



- (I) State whether  $N$  and  $E_k$  of gas B are larger than, smaller than or the same as the corresponding values of those of gas A found in (a)(i). (2 marks)
- (II) Given that the root-mean-square speed ( $c_{\text{rms}}$ ) of the molecules of gas A is  $600 \text{ m s}^{-1}$ , estimate  $c_{\text{rms}}$  of the molecules of gas B. (2 marks)
- (b) Figure 2.3 shows two glass jars containing air and gas C respectively and they are separated by a glass plate. Both are at the same pressure and temperature. Gas C is red in colour.



After removing the glass plate, it takes a few minutes for gas C to diffuse several centimetres into the upper jar despite its molecules having a root-mean-square speed of  $200 \text{ m s}^{-1}$ . Explain this observation. (2 marks)

**(a)(i)(I)** From  $pV = nRT$ ,  $n = (1.0 \times 10^5)(6.0 \times 10^{-4})/[8.31(300)] = 0.0241$  mol. Hence  $N = nN_A = 1.45 \times 10^{22}$  molecules.

(a)(ii)(I)  $N$  is unchanged because  $p$ ,  $V$  and  $T$  are unchanged.  $E_k$  is unchanged because the temperature is unchanged.

**(b)** Gas molecules repeatedly collide with air molecules, continually changing direction. Their zig-zag paths are much longer than their net upward displacement, so diffusion over a few centimetres takes much longer than distance divided by molecular speed.

### Question 3 — Forces and circular motion

Figure 3.1 shows a quadcopter with four propellers. When the four propellers are operating to propel air streams vertically downwards, the quadcopter can hover in the air in a fixed position. ( $g = 9.81 \text{ m s}^{-2}$ )



(a) Referring to Newton's laws of motion, explain why the quadcopter can hover in the air. (2 marks)

Given: mass of the quadcopter = 1.38 kg; total area swept by the four propellers =  $0.284 \text{ m}^2$ ; density of air =  $1.20 \text{ kg m}^{-3}$ .

(b) Suppose the speed of the air streams produced is  $v$ .

(i) Considering the total volume of air being propelled downwards in 1 second, express the mass of air propelled by the quadcopter per second in terms of  $v$ . (2 marks)

(ii) Hence, find the speed  $v$  that enables the quadcopter to hover. (2 marks)

(c) The quadcopter can be adjusted to tilt at an angle  $\theta$  to the vertical as shown in Figure 3.2(a) and fly along a horizontal circular path of radius  $r$  [Figure 3.2(b)]. Neglect the size of the quadcopter and air resistance in your calculation.

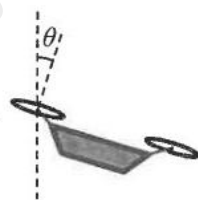


Figure 3.2(a)

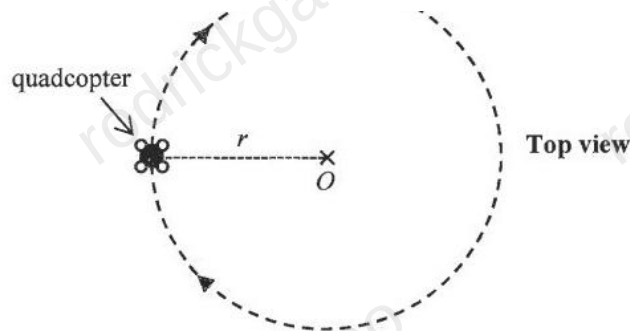


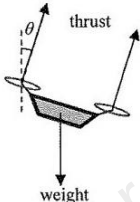
Figure 3.2(b)

(i) On Figure 3.2(a), indicate and label the force(s) acting on the quadcopter. (2 marks)

\* (ii) Find the centripetal force required for the quadcopter to follow such a circular path with a radius of 50 m and a speed of  $15 \text{ m s}^{-1}$ . (2 marks)

(iii) Hence, calculate the angle  $\theta$  to attain such centripetal force for the quadcopter. (2 marks)

## Official marking scheme

| Solution                                                                                                                                                                                                                                                                                                                     | Marks         | Remarks                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3. (a) According to Newton's third law of motion, force on air streams and the thrust on the quadcopter are (equal and) opposite, with magnitude given by $F = \text{rate of change of momentum (of air streams)}$ . Therefore the thrust (upward) on the quadcopter balances its weight for a certain speed of air streams. | 1A<br>1A<br>2 |                                                                                                                                                                                   |
| (b) (i) Volume of air streams propelled in 1 s<br>$V = 0.284 v$<br><br>$m_a = \text{density} \times \text{volume}$<br>$= 1.20 \times 0.284 v = 0.3408 v$                                                                                                                                                                     | 1M<br>1A<br>2 |                                                                                                                                                                                   |
| (ii) Weight of quadcopter = rate of change of momentum of air streams<br>$1.38 g = m_a v - 0$<br>$1.38 \times 9.81 = (0.3408 v) \times v$ (From (b)(i))<br>$v^2 = 39.723592$<br>$v = 6.302665 \text{ m s}^{-1} \approx 6.30 \text{ m s}^{-1}$                                                                                | 1M<br>1A<br>2 | For $g = 10 \text{ m s}^{-2}$ ,<br>$v = 6.363408 \text{ m s}^{-1} \approx 6.36 \text{ m s}^{-1}$                                                                                  |
| (c) (i)                                                                                                                                                                                                                                     | 2A<br>2       |                                                                                                                                                                                   |
| (ii) centripetal force required<br>$F = \frac{1.38 \times 15^2}{50}$<br>$= 6.21 \text{ N}$                                                                                                                                                                                                                                   | 1M<br>1A<br>2 |                                                                                                                                                                                   |
| (iii) Let $T = \text{total thrust on the quadcopter}$<br>Vertical: $T \cos \theta = 1.38 g$ ( $g = 9.81 \text{ m s}^{-2}$ )<br>Horizontal: $T \sin \theta = 6.21 \text{ N}$ (From (c)(ii))<br>On solving the two equations,<br>$\tan \theta = 0.458716$<br>$\theta = 24.641662^\circ \approx 24.6^\circ$                     | 1M<br>1A<br>2 | Accept $\tan \theta = \frac{\text{centripetal force}}{\text{weight}}$<br>For $g = 10 \text{ m s}^{-2}$ ,<br>$\tan \theta = 0.45$<br>$\theta = 24.227745^\circ \approx 24.2^\circ$ |

(a) The propellers exert a downward force on the air. By Newton's third law, the air exerts an equal upward thrust on the quadcopter. At the hovering speed, this thrust equals the weight, giving zero resultant force and zero acceleration.

(b)(i) The volume moved per second is  $\Delta V = 0.284v$ . The mass flow rate is  $\rho \Delta V = 1.20(0.284)v = 0.3408v \text{ kg s}^{-1}$ , with  $v$  in  $\text{m s}^{-1}$ .

(b)(ii) Thrust equals the rate of downward momentum transfer. For hovering,  $mg = (\rho \Delta V)v$ , so  $1.38(9.81) = 0.3408v^2$  and  $v = 6.30 \text{ m s}^{-1}$ .

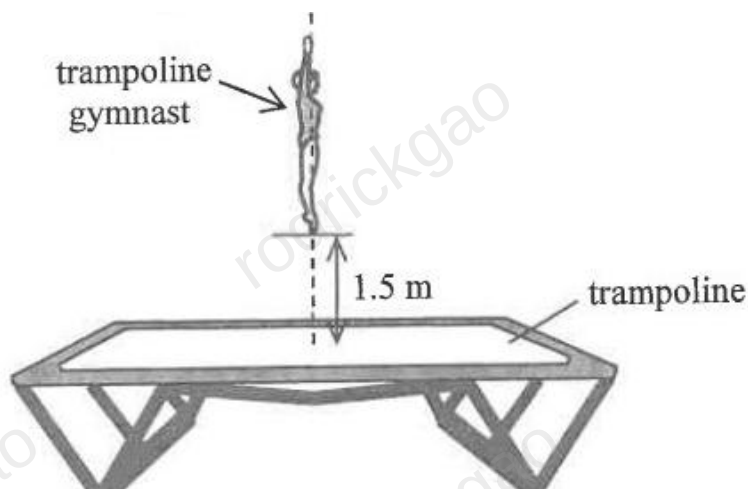
(c)(i) Draw weight  $mg$  vertically downward and total thrust  $T$  perpendicular to the propeller plane, tilted towards the centre of the circular path. The centripetal force is the horizontal resultant, not an additional force.

(c)(ii)  $F_c = mv^2/r = 1.38(15^2)/50 = 6.21 \text{ N}$  towards the centre.

(c)(iii)  $T \cos \theta = mg$  and  $T \sin \theta = F_c$ . Thus  $\tan \theta = 6.21/[1.38(9.81)]$ , giving  $\theta = 24.6^\circ$ .

### Question 4 — Energy

Figure 4.1 shows a trampolene gymnast of mass 50 kg performing a straight jump. Her feet are 1.5 m above the trampolene at the maximum height. Neglect air resistance and assume that the gymnast maintains this posture throughout the jump. ( $g = 9.81 \text{ m s}^{-2}$ )



- (a) Find the kinetic energy of the gymnast just as her feet touch the trampolene on the way down from her jump. (2 marks)
- (b) After touching the trampolene, the gymnast keeps on moving downward for 0.40 m further before she stops.
  - (i) Describe the energy transfer to the trampolene by the gymnast after touching the trampolene. (2 marks)
  - (ii) Estimate the average force exerted by the gymnast on the trampolene during this 0.40 m downward movement. (2 marks)

## Official marking scheme

| Solution                                                                                                                                                                                     | Marks         | Remarks                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------|
| 4. (a) $\frac{1}{2}mv^2 = mgh$<br>$= (50)(9.81)(1.5)$<br>$= 735.75 \text{ J} \approx 736 \text{ J}$                                                                                          | 1M<br>1A<br>2 | For $g = 10 \text{ m s}^{-2}$ , K.E. = 750 J               |
| (b) (i) Kinetic energy and potential energy (of the athlete) change to elastic potential energy (of the trampoline).                                                                         | 2A<br>2       |                                                            |
| (ii) $\bar{F}d = mgh + mgd$ [Or $\bar{F}d = \frac{1}{2}mv^2 + mgd$ ]<br>$\bar{F} = \frac{50(9.81)(1.5+0.40)}{0.40}$<br>$= 1839.375 + 490.5$<br>$= 2329.875 \text{ N} \approx 2330 \text{ N}$ | 1M<br>1A      | For $g = 10 \text{ m s}^{-2}$ , $\bar{F} = 2375 \text{ N}$ |
| Or<br>$\bar{F} = \frac{1}{0.40} [735.75 + (50)(9.81)(0.40)]$<br>$\approx 2330 \text{ N}$                                                                                                     | 1M<br>1A      |                                                            |
|                                                                                                                                                                                              | 2             |                                                            |

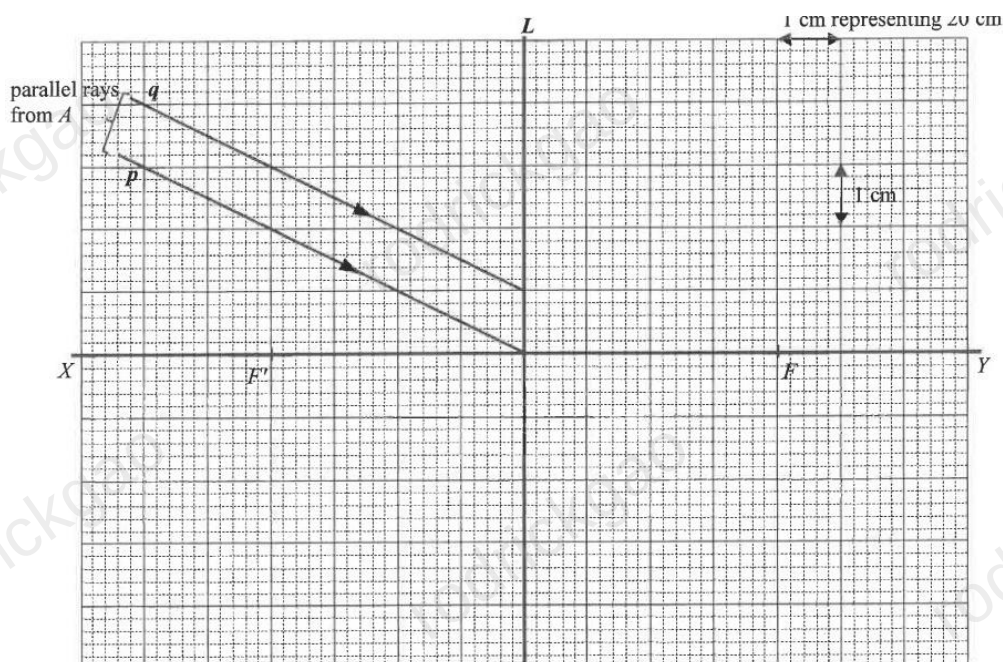
(a) Before contact, loss of gravitational potential energy becomes kinetic energy:  $K = mgh = 50(9.81)(1.5) = 736 \text{ J}$ .

(b)(i) The gymnast's kinetic energy and the further loss of gravitational potential energy are transferred into elastic potential energy of the trampoline.

(b)(ii) During the additional downward displacement  $d$ ,  $F_{av} d = K + mgd = mg(h+d)$ . Thus  $F_{av} = 50(9.81)(1.5+0.40)/0.40 = 2.33 \times 10^3 \text{ N}$ . This is the magnitude of the gymnast's downward force on the trampoline; using  $K/d$  alone would omit the work done by gravity during compression.

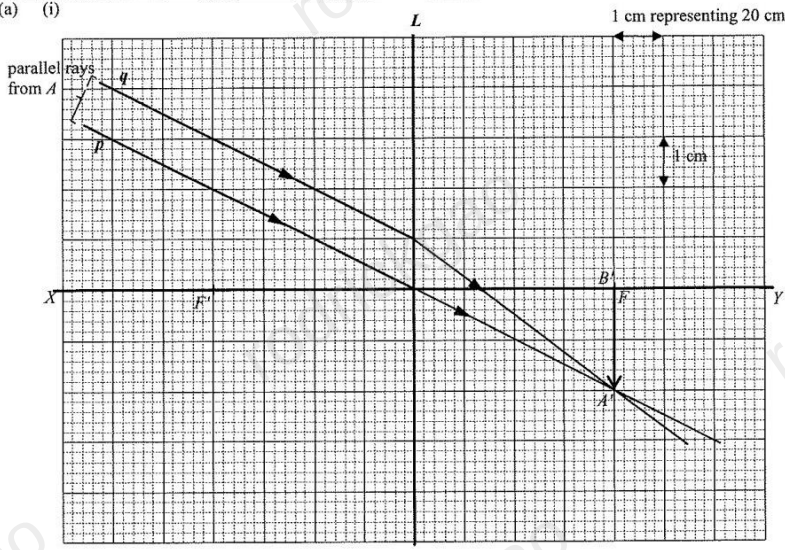
## Question 5 — Geometrical optics

In the figure below, XY is the horizontal principal axis of a convex lens L with principal foci F and F'. p and q are parallel rays coming from point A of a distant object AB. (The object can be represented by a vertical arrow A/B, but it is not shown on the figure and its end B is on the principal axis.)



- (a) (i) Draw the refracted rays of p and q so as to locate the image of A (denoted as A'). Hence mark the image A'B' of object AB. (3 marks)
- (ii) Suggest an experiment to verify whether a real image is formed in the above situation. (2 marks)
- (b) (i) Use the ray diagram drawn to estimate the ratio: height of object AB / distance of AB from L. The horizontal and vertical scales are 1:20 and 1:1 respectively. (2 marks)
- (ii) Hence, estimate the height of object AB, which is a lamp post at a distance of 200 m from lens L. (1 mark)

## Official marking scheme

| Solution                                                                                                                                                                                                                                                                                                         | Marks                         | Remarks                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------|
| <p>5. (a) (i)</p>  <p>refracted rays of <math>p</math>, <math>q</math> correctly drawn and <math>A'</math> correctly marked.<br/>image <math>A'B'</math> correctly marked.</p>                                                 | <p>1A<br/>1A<br/>1A<br/>3</p> | <p>1 cm representing 20 cm</p>                                       |
| <p>(ii) Image <math>A'B'</math> of a distant object, say, a building is real and therefore it can be captured by a screen (placed at the focal plane).</p>                                                                                                                                                       | <p>1A<br/>1A<br/>2</p>        |                                                                      |
| <p>(b) (i) Ratio by similar triangles,<br/> <math display="block">\frac{\text{height of } AB}{\text{object distance}} = \frac{\text{height of } A'B'}{\text{focal length / image distance}}</math> <math display="block">= \frac{2}{4 \times 20} = \frac{1}{40}</math> <math display="block">= 0.025</math> </p> | <p>1M<br/>1A<br/>2</p>        | <p>Accept height of image <math>A'B'</math>:<br/>1.8 cm ~ 2.2 cm</p> |
| <p>(ii) <math>\frac{\text{height of } AB}{\text{object distance}} = \frac{1}{40} = 0.025</math><br/>           Height of <math>AB = 0.025 \times 200 = 5</math> m         </p>                                                                                                                                   | <p>1A<br/>1</p>               |                                                                      |

**(a)(i)** Extend ray  $p$  undeviated through the optical centre. Its intersection with the focal plane through the right-hand focus  $F$  locates  $A'$ . Draw refracted ray  $q$  through  $A'$ . Put  $B'$  at  $F$  and draw the inverted image arrow from  $B'$  to  $A'$ .

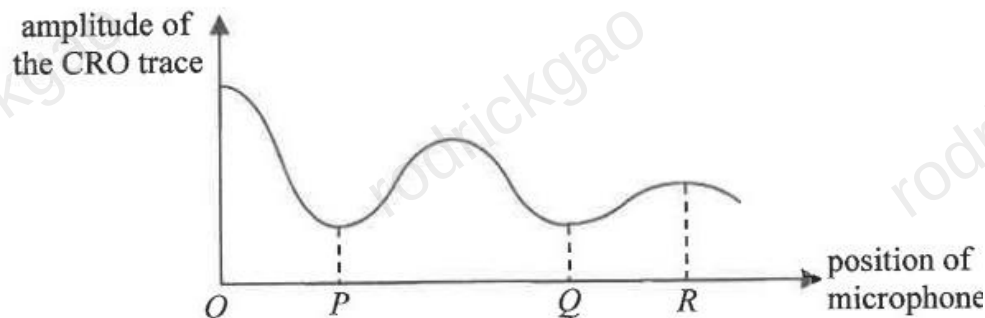
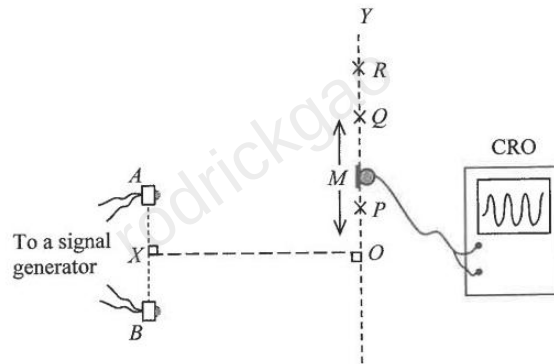
**(a)(ii)** Place a screen behind the lens near the focal plane and adjust its position. A sharp image that can be captured on the screen confirms that the image is real.

**(b)(i)** Similar triangles give object height/object distance = image height/image distance. On the grid, the drawn image height is about half the horizontal focal distance. Accounting for the horizontal 1:20 and vertical 1:1 scales gives ratio =  $0.5/20 = 0.025 = 1/40$ .

**(b)(ii)** Height =  $0.025(200 \text{ m}) = 5 \text{ m}$ . Values within the permitted diagram-reading tolerance are acceptable.

## Question 6 — Interference of sound

In Figure 6.1, two small identical loudspeakers A and B produce coherent sound waves. X is the mid-point of AB. A microphone M connected to a CRO is moved along OY to detect the loudness of the sound, with a CRO trace of larger amplitude representing greater loudness. Figure 6.2 shows the result.



- Explain what is meant by coherent sound waves. (1 mark)
- Explain why sound of alternate maximum and minimum loudness is detected along OY. (2 marks)
  - The amplitude of the CRO trace at P is not zero. Suggest a possible reason. (1 mark)
- Given:  $AQ = 2.17$  m,  $BQ = 2.58$  m. Find the speed of sound in air if the frequency of the signal generator is 1200 Hz. (2 marks)
- Given that the separation of A and B is 0.80 m, explain why no more maximum can be detected when the microphone is moved beyond position R along OY. (2 marks)
- The microphone is now moved from O to X along the line OX. State whether the amplitude of the CRO trace increases, decreases, remains the same, or varies periodically. (1 mark)

## Official marking scheme

| Solution                                                                                                                                                                                                                                                                                                                                                                                     | Marks | Remarks                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------|
| 6. (a) Sound waves having the same frequency / wavelength and a constant phase difference (or always in phase / in opposite phase) are coherent.                                                                                                                                                                                                                                             | 1A    |                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                              | 1     |                                                           |
| (b) (i) When the sound waves from <i>A</i> and <i>B</i> meet at various positions along <i>OY</i> , interference occurs.<br>At positions where the two waves are in phase, constructive interference occurs and gives maximum (loudness).<br>At positions where the two waves are in opposite phase, destructive interference occurs and gives minimum (loudness).                           | 1A    |                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                              | 1A    |                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                              | 2     |                                                           |
| (ii) Any ONE:<br>• due to background noise<br>• due to reflection of unwanted sound from the wall, floor etc.<br>• the intensity of sound waves from <i>A</i> and <i>B</i> reaching <i>P</i> are NOT equal as $AP < BP$<br>therefore cancellation is incomplete.                                                                                                                             | 1A    |                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                              | 1     |                                                           |
| (c) Path difference at <i>Q</i><br>$3\lambda/2 = 2.58 - 2.17$<br>$= 0.41 \text{ m}$<br>$\therefore \lambda = 0.2733333 \text{ m} \approx 0.273 \text{ m}$<br>$v = f\lambda$<br>$= (1200)(0.273)$<br>$= 328 \text{ m s}^{-1}$                                                                                                                                                                 | 1M    |                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                              | 1A    | Accept $327.6 \text{ m s}^{-1}$ to $328 \text{ m s}^{-1}$ |
|                                                                                                                                                                                                                                                                                                                                                                                              | 2     |                                                           |
| (d) Path difference $\Delta$ at any position along <i>OY</i> from <i>A</i> and <i>B</i> is less than <i>AB</i> (i.e. $\Delta = n\lambda < AB$ )<br>max. $\Delta = 0.80 \text{ m} = \frac{0.8}{0.273} \lambda \approx 2.93 \lambda$<br>i.e. max. $\Delta < 3 \lambda$<br>As path difference cannot be equal to $3 \lambda, 4 \lambda, \dots$ ,<br>therefore no more maximum beyond <i>R</i> . | 1M    |                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                              | 1A    |                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                              | 2     |                                                           |
| (e) increases                                                                                                                                                                                                                                                                                                                                                                                | 1A    |                                                           |
|                                                                                                                                                                                                                                                                                                                                                                                              | 1     |                                                           |

(a) Coherent sound waves have the same frequency and a constant phase difference.

(b)(i) The waves superpose. Where they arrive in phase, constructive interference gives a maximum; where they arrive in opposite phase, destructive interference gives a minimum. The changing path difference along *OY* produces alternating maxima and minima.

(b)(ii) At *P*, the waves from the two speakers can have unequal amplitudes because their travel distances differ, so cancellation is incomplete. Background noise or reflected sound is another valid reason.

(c) *Q* is the second minimum, so  $BQ - AQ = 3\lambda/2 = 2.58 - 2.17 = 0.41 \text{ m}$ . Thus  $\lambda = 0.2733 \text{ m}$  and  $v = f\lambda = 1200(0.2733) = 328 \text{ m s}^{-1}$ .

(d) At any point, the path difference is less than the speaker separation,  $0.80 \text{ m}$ . The next maximum beyond *R* would require  $3\lambda = 0.820 \text{ m}$ , which is larger than  $0.80 \text{ m}$ , so it cannot occur. Higher-order maxima are also impossible.

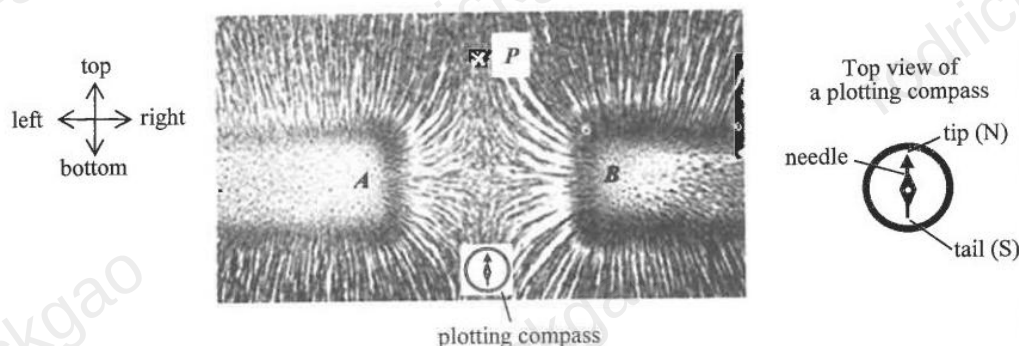
(e) The amplitude increases. Along *OX* the distances from *A* and *B* remain equal, so the waves reinforce throughout, and both waves become stronger as the microphone approaches the speakers.

## Question 7 — Magnetic fields

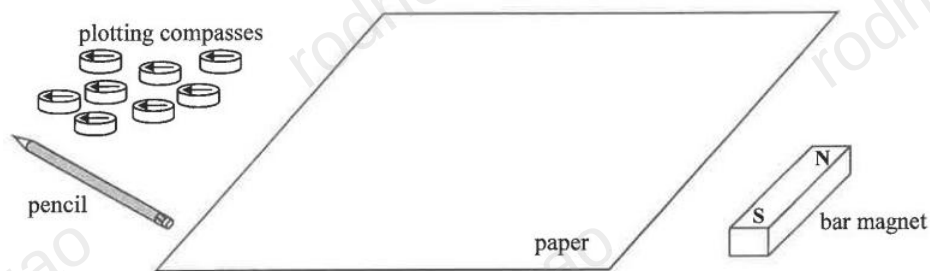
Read the following passage about magnetic field patterns and answer the questions that follow.

Iron filings are tiny pieces of iron nearly in powder form. Since iron is ferromagnetic, a magnetic field would induce each piece of iron filings to become a ‘tiny bar magnet’. The south pole of each of these ‘tiny bar magnets’ attracts the north poles of neighbouring iron filings. The magnetic field pattern is displayed as iron filings align themselves with the field lines.

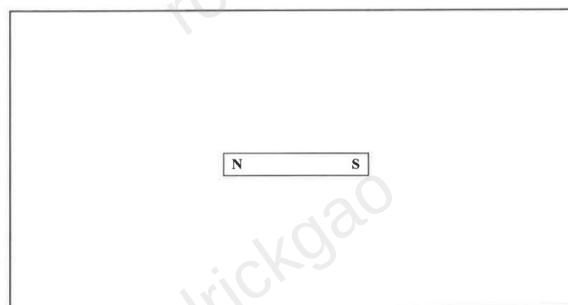
The figure below shows such a pattern displayed on a cardboard with two identical bar magnets placed underneath. A plotting compass placed at the bottom of the figure points to the top as shown.



- (a) (i) State the polarity of the respective poles at A and B of the bar magnets. (1 mark)
  - (ii) If the compass is moved to P, towards what direction (top, bottom, left or right) would it be pointing? (1 mark)
  - (iii) In obtaining such field patterns experimentally, one is advised to place the magnets underneath the cardboard. Why? (1 mark)
- (b) You are given a bar magnet (with polarities marked), 8 small plotting compasses, a pencil and a piece of white paper as shown in Figure 7.1.

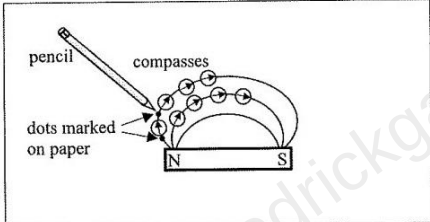


- (i) Describe, with the aid of a diagram, how you would use the apparatus given to trace several field lines around the bar magnet. Neglect the Earth's magnetic field. (5 marks)



- (ii) Suggest ONE advantage of using a plotting-compass method over an iron-filing method in studying magnetic fields. (1 mark)

### Official marking scheme

| Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks    | Remarks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|
| 7. (a) (i) A: south pole (S)<br>B: south pole (S)                                                                                                                                                                                                                                                                                                                                                                                                               | 1A<br>1  |         |
| (ii) (Towards) bottom / downwards                                                                                                                                                                                                                                                                                                                                                                                                                               | 1A<br>1  |         |
| (iii) Iron filings come into physical contact with the magnet<br>will never come off / are difficult to get rid of.                                                                                                                                                                                                                                                                                                                                             | 1A<br>1  |         |
| (b) (i) Successively place the compasses on the paper near one<br>pole (say N) of the magnet such that the tip of each<br>compass needle follows / lines up with the tail of the<br>compass ahead.<br>Diagram                                                                                                                                                                                                                                                   | 1A<br>1A |         |
|                                                                                                                                                                                                                                                                                                                                                                                |          |         |
| Denote the tip and tail (of each compass) with dots on the<br>paper using the pencil or remove the compasses one by one<br>and trace the (direction of) field line.                                                                                                                                                                                                                                                                                             | 1A       |         |
| Draw / sketch a smooth curve representing a field line by<br>joining up the series of dots/segments going from one pole<br>to the other.                                                                                                                                                                                                                                                                                                                        | 1A       |         |
| Repeat the above process by starting from different points<br>around the magnet to get another/several field line(s).                                                                                                                                                                                                                                                                                                                                           | 1A<br>5  |         |
| (ii) Any ONE:<br><ul style="list-style-type: none"> <li>Compass is sensitive enough to explore very weak magnetic field such as the Earth's field.</li> <li>The drawing can show the direction of the field lines / the drawing of the field will still be on the paper after removing the magnet.</li> <li>It is difficult to draw individual field line separately in iron-filing method even if the filings are sprinkled very thinly and evenly.</li> </ul> | 1A<br>1  |         |

**(a)(i)** Both A and B are south poles. Field lines outside a magnet enter its south pole; use the given compass direction to establish the field direction.

**(a)(ii)** The compass points towards the bottom of the figure.

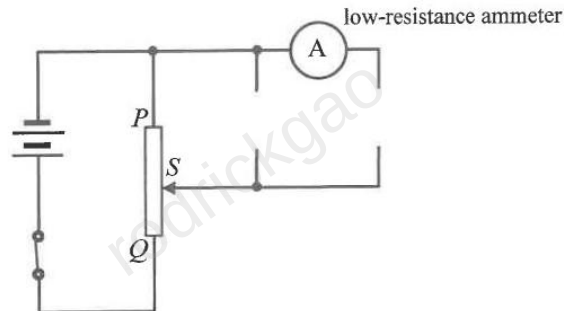
**(a)(iii)** The cardboard prevents the iron filings from touching and sticking to the magnets, from which they would be difficult to remove.

**(b)(i)** Place the magnet on the paper. Arrange the plotting compasses successively along a field line, with neighbouring needles aligned head to tail, as in the official diagram. Mark each needle's two ends with pencil dots. Remove the compasses and join the dots with a smooth curve, adding arrows from N towards S outside the magnet. Repeat from different starting points near the magnet to trace several field lines. Include the magnet, compass positions, dots and direction arrows in the sketch.

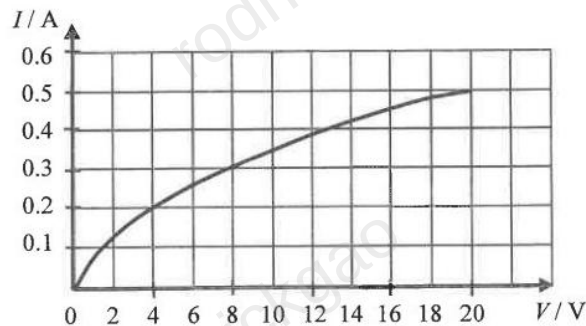
**(b)(ii)** A plotting compass reveals the direction of the magnetic field; iron filings show the pattern but do not distinguish its direction.

## Question 8 — Current–voltage characteristics

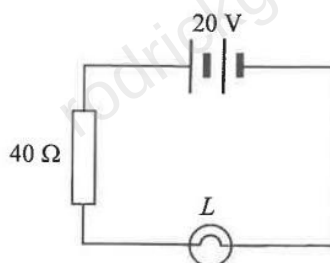
- (a) A student sets up the circuit in Figure 8.1 to find the current–voltage (I–V) characteristic of a filament light bulb. PQ is a variable resistor in the circuit with sliding contact S. The bulb and a high-resistance voltmeter are missing in the circuit.



- (i) Use suitable circuit symbols to complete the circuit. (1 mark)
- (ii) How would the brightness of the light bulb change when contact S is adjusted from P to Q? (1 mark)
- (b) The graph below represents the I–V characteristic of this light bulb, which has a rated voltage of 20 V.

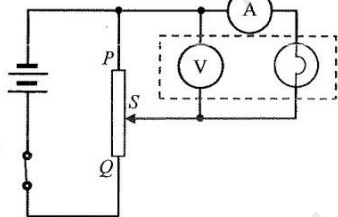
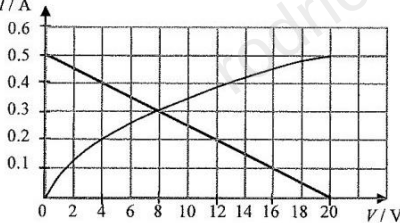


- (i) Find the resistance of the bulb when working under its rated voltage. (2 marks)
- (ii) Explain why the resistance of the bulb varies with the applied voltage V. (2 marks)
- (c) The bulb L in (b) and a  $40\ \Omega$  resistor are connected in series with a 20 V battery of negligible internal resistance as shown in Figure 8.3. The current I in the circuit and the voltage V across the bulb are related by  $I = 0.5 - 0.025V$ .



- (i) Add a straight line on Figure 8.2 to determine the current I. (2 marks)
- (ii) Hence estimate the power consumed by bulb L. (2 marks)

## Official marking scheme

| Solution                                                                                                                                                                                  | Marks    | Remarks                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------|
| 8. (a) (i)                                                                                               | 1A       |                                                        |
| (ii) Brightness increases.                                                                                                                                                                | 1A       |                                                        |
| (b) (i) Resistance = $\frac{V}{I} = \frac{20}{0.5} = 40 \Omega$                                                                                                                           | 1M<br>1A |                                                        |
| (ii) As the applied voltage $V$ increases, current $I$ / electrical power increases and this raises the filament's temperature, thus the resistance $R$ of the filament / bulb increases. | 1A<br>1A |                                                        |
| Or<br>the electrons collide with the increasingly vibrating atoms / lattice ions (of the filament) which impede their flow, i.e. $R$ increases.                                           | 1A       |                                                        |
| (c) (i) <br>Current = 0.3 A (at $V = 8$ V)                                                              | 1M<br>1A | Correct straight line intersects the curve at (8, 0.3) |
| (ii) Power consumed = $IV$<br>= (0.3)(8)<br>= 2.4 W                                                                                                                                       | 1M<br>1A | e.c.f. from (c)(i)                                     |

(a)(i) Connect the bulb in series with the ammeter in the right-hand branch. Connect the high-resistance voltmeter in parallel across the bulb, as shown in the official circuit.

(a)(ii) Brightness increases: moving S from P towards Q increases the voltage across the bulb.

(b)(i) At the rated voltage, the graph gives  $I = 0.50$  A. Thus  $R = V/I = 20/0.50 = 40 \Omega$ .

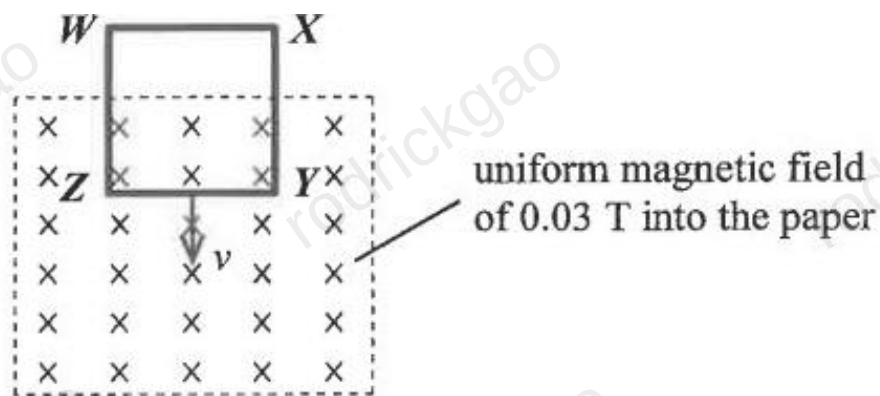
(b)(ii) A higher voltage increases the electrical power and filament temperature. Increased lattice vibrations cause more frequent scattering of conduction electrons, so the filament resistance increases.

(c)(i) Plot the load line  $I = 0.5 - 0.025V$  through (0 V, 0.5 A) and (20 V, 0 A). Its intersection with the bulb curve is approximately (8 V, 0.30 A), so the current is 0.30 A.

(c)(ii) Power in the bulb is  $P = VI = 8(0.30) = 2.4$  W.

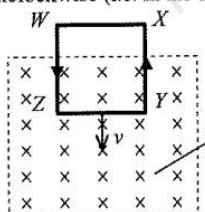
### Question 9 — Electromagnetic induction

A square metal loop WXYZ of side length 0.10 m is inserted with a constant velocity  $v$  into a uniform magnetic field of flux density 0.03 T. The magnetic field is perpendicular to the plane of the loop as shown in Figure 9.1. The resistance of each side of the metal loop is  $0.15\ \Omega$ . When the metal loop is entering the magnetic field, a current of 0.01 A flows in the loop.



- (a) On Figure 9.1, indicate the direction of this current. (1 mark)
- \* (b) Find  $v$ . (2 marks)
- (c) (i) Find the potential difference  $V_{YZ}$  between Y and Z. (2 marks)
- (ii) Explain whether  $V_{YZ}$  is equal to the induced e.m.f. across YZ. (1 mark)

## Official marking scheme

| Solution                                                                                                                                                                                                                       | Marks    | Remarks                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------|
| 9. (a) Induced current is anticlockwise (i.e. in the direction ZYXW)<br> uniform magnetic field of 0.03 T into the paper                      | 1A       |                                                                              |
|                                                                                                                                                                                                                                | 1        |                                                                              |
| (b) Current $I = \frac{\varepsilon}{R} = \frac{Blv}{R}$ [Or $\varepsilon = \frac{N\Delta\Phi}{\Delta t} = \frac{B\Delta A}{\Delta t} = Blv = IR$ ]<br>$0.01 = \frac{0.03(0.10)v}{4 \times 0.15}$<br>$v = 2.0 \text{ m s}^{-1}$ | 1M<br>1A |                                                                              |
| Or<br>Power input $P = Fv = I^2 R$<br>$(IlB)v = I^2 R$<br>$Blv = IR$<br>$(0.03)(0.10)v = (0.01)(4 \times 0.15)$<br>$v = 2.0 \text{ m s}^{-1}$                                                                                  | 1M<br>1A |                                                                              |
|                                                                                                                                                                                                                                | 2        |                                                                              |
| (c) (i) $V_{YZ} = I(R_{ZW} + R_{WX} + R_{XY})$<br>$= 0.01(0.15 + 0.15 + 0.15)$<br>$= 0.0045 \text{ V} (= 4.5 \text{ mV})$                                                                                                      | 1M<br>1A |                                                                              |
| Or<br>$V_{YZ} = \varepsilon - IR_{YZ}$<br>$= Blv - IR_{YZ}$<br>$= 0.03(0.10)(2.0) - 0.01(0.15)$<br>$= 0.0045 \text{ V}$                                                                                                        | 1M<br>1A |                                                                              |
|                                                                                                                                                                                                                                | 2        |                                                                              |
| (ii) As there is voltage drop ( $IR_{YZ}$ ) within / across YZ,<br>$V_{YZ}$ is less than the induced e.m.f. across YZ ( $\varepsilon = Blv$ )                                                                                  | 1A       | Note:<br>$\varepsilon - IR_{YZ} = V_{YZ}$<br>$\varepsilon = 0.006 \text{ V}$ |
|                                                                                                                                                                                                                                | 1        |                                                                              |

(a) The current is anticlockwise,  $Z \rightarrow Y \rightarrow X \rightarrow W \rightarrow Z$ . The magnetic flux into the paper increases as the loop enters; by Lenz's law the induced current creates a field out of the paper.

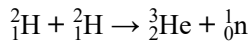
(b) Total resistance  $R = 4(0.15) = 0.60 \, \Omega$ . The induced e.m.f. is  $\varepsilon = Blv = IR$ , so  $v = 0.01(0.60)/[0.03(0.10)] = 2.0 \text{ m s}^{-1}$ .

(c)(i) The potential difference between Y and Z equals the drop across the other three sides:  $V_{YZ} = I(3 \times 0.15) = 0.0045 \text{ V} = 4.5 \text{ mV}$ .

(c)(ii) It is smaller than the induced e.m.f.,  $\varepsilon = Blv = 6.0 \text{ mV}$ , because side YZ itself has a resistive drop  $IR_{YZ} = 1.5 \text{ mV}$ . Hence  $V_{YZ} = \varepsilon - IR_{YZ}$ .

## Question 10 — Nuclear energy

Deuterium ( ${}^2_1\text{H}$ ) and tritium ( ${}^3_1\text{H}$ ) are isotopes of hydrogen. The fusion of two deuterium nuclides can be represented by the following equation:



Given: mass of  ${}^2_1\text{H} = 2.014102 \text{ u}$ ; mass of  ${}^3_2\text{He} = 3.016029 \text{ u}$ ; mass of  ${}^1_0\text{n} = 1.008665 \text{ u}$ .

\* (a) 1 in 6420 atoms of hydrogen in nature is a deuterium. Estimate the maximum energy, in MeV, that could be produced from this fusion reaction with 1 mole of hydrogen nuclides. (3 marks)

(b) If conditions are altered, the fusion of two deuterium nuclides may not produce a helium (He) nucleus.

Complete the equation below for such a possible fusion reaction. (1 mark)



(c) Both fission and fusion produce energy. State TWO advantages of using fusion as an energy source over fission. (2 marks)

### Official marking scheme

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                         |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 10. (a) | Decrease in mass in the reaction (for two ${}^2_1\text{H}$ )<br>$\Delta m = (2 \times 2.014102) - (3.016029 + 1.008665)$<br>$= 4.028204 - 4.024694 = 0.003510 \text{ u}$<br>Maximum energy released by 1 mole of hydrogen nuclides<br>$= \left( \frac{6.02 \times 10^{23}}{6420 \times 2} \right) \times (0.003510 \times 931 \text{ MeV})$<br>$= 1.532104 \times 10^{20} \text{ (MeV)} \approx 1.53 \times 10^{20} \text{ (MeV)}$ | 1M<br><br>1M<br>1A<br>3 |
| (b)     | ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^3_1\text{H} + {}^1_1\text{H}$ (or ${}^3_1\text{H}$ )                                                                                                                                                                                                                                                                                                                               | 1A<br>1                 |
| (c)     | Any TWO:<br>- Unlike fission, the fuel (hydrogen) for fusion is abundant and widely available.<br>- Unlike fission, the products of fusion reactions are not radioactive.<br>- Much larger amount of energy is produced by fusion (for equal mass of fuel).                                                                                                                                                                        | 2A<br><br>2             |

(a) The mass defect per reaction is  $\Delta m = 2(2.014102) - 3.016029 - 1.008665 = 0.003510 \text{ u}$ . Each reaction releases  $0.003510(931) = 3.268 \text{ MeV}$ . One mole of natural hydrogen contains  $N_A/6420$  deuterium nuclides, and each reaction consumes two. Total energy =  $[6.02 \times 10^{23} / (6420 \times 2)](0.003510)(931) = 1.53 \times 10^{20} \text{ MeV}$ .

(b) The alternative reaction is  ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^3_1\text{H} + {}^1_1\text{H}$ . Both nucleon number and proton number are conserved.

(c) Two advantages are the abundance and wide availability of hydrogen fuel, and the greater energy release per unit mass of fuel for the fusion reactions considered. Avoid the blanket claim that every fusion reaction produces no radioactive material: tritium in part (b) is radioactive.